Paper 36: Nash Equilibrium in Autonomous Agent Labor Market

Pillar: Network Economics Distributions — Autonomous Research Ecosystem

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Abstract

We hypothesize that Nash Equilibrium in Autonomous Agent Labor Market achieves Stable market clear on Python / Rust (Double Auction) while preserving strict mathematical invariants across edge, creative graph-loop, and decentralized clearing layers. We present a formal mathematical model, a production implementation in `Python / Rust (Double Auction)`, and an automated empirical verification harness targeting **Stable market clear**. All benchmarks and invariants are reproducibly verified via our continuous Research Graph-Loop pipeline [3, 6, 2].

1 Introduction & Formal Hypothesis

Modern autonomous agent architectures face fundamental bottlenecks across constrained edge silicon, deterministic multi-modal synthesis, and decentralized economic clearing. In this work (Paper 36 within the *Network Economics Distributions* pillar), we address these limitations directly.

Core Hypothesis: We hypothesize that Nash Equilibrium in Autonomous Agent Labor Market achieves Stable market clear on Python / Rust (Double Auction) while preserving strict mathematical invariants across edge, creative graph-loop, and decentralized clearing layers.

2 Mathematical Formulation

We formulate our system objective and state transition invariant as:

$$\mathcal{L}_{\text{total}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} \left[\mathcal{D}_{\text{KL}}(P_{\text{target}} \parallel P_{\text{draft}}) \right] + \lambda \| \Delta_{\text{state}} \|_2^2 \tag{1}$$

Subject to strict runtime verification bounds (`Stable market clear`) executed across our hybrid compute fabric (Local Edge Silicon, Homelab Silo Server, and FAU Research Computing HPC nodes) [4, 5, 1].

3 System Architecture & Implementation

The reference architecture is implemented in `Python / Rust (Double Auction)` across the following work package modules:

- `crates/agent_market/src/lib.rs`
- `research_ecosystem/pillar3_economics.py`

4 Empirical Evaluation & Benchmarks

We executed the automated verification suite for Paper 36 and recorded the following empirical metrics (Table 1):

Table 1: Empirical Verification Results for Paper 36

Metric / Invariant	Measured Value
matched_trades	3
clearing_price	8.0
social_welfare	13.0
passed	True

5 Conclusion

We have demonstrated that *Nash Equilibrium in Autonomous Agent Labor Market* satisfies its target specification (`Stable market clear`) with deterministic reproducibility.

References

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